

THE AGE OF discovery

SCIENCE AND PROGRESS

The 19th century was a time of immense change. In 1800, most people traveled by foot, and communication was by letter. Yet in the hundred years that followed, new scientific discoveries came one after another, and by the 1900s, large swathes of the world were linked together by railroads and telegraph wires. Internal combustion engines powered vehicles, microbes were proved to cause diseases, electricity was harnessed and utilized, and the theory of evolution was proposed. In 1833, the term “scientist” was coined to describe the researchers who were changing the world—a growing number of whom were women.

One of the first female scientists of the age was Scottish polymath Mary Somerville, who studied math and astronomy. Another pioneer was Ada Lovelace, who worked with Charles Babbage on his Analytical Engine, an early version of the computer. She became an expert in sequencing instructions on the punched cards that it used, and wrote the earliest machine algorithm, becoming the world’s first computer programmer. Perhaps the most famous female scientist of the age, however, was Polish chemist and physicist Marie Curie (see right), who studied radioactivity.

Outside of the laboratory, in the 19th century naturalists revolutionized the study of living things. Breaking with religious teachings about creation, paleontologists including Mary Anning and Elizabeth Philpot dug fossils out of the earth and used them to piece together a picture of prehistoric life. Biologists such as Jeanne Villepreux-Power, who invented the aquarium, studied living animals. The botanist Anna Atkins merged the study of the natural world with cutting-edge technology, using the new invention of photography to make images of algae and plants. In the medical world, Elizabeth Blackwell and Elizabeth Garrett Anderson fought for, and won, the right to train and qualify as doctors in the US and Britain respectively.

Making the modern world

Women were also involved in engineering and inventing. When her engineer husband became incapacitated, Emily Warren Roebling took over the management of the construction of the Brooklyn Bridge. In Britain, inventor Hertha Marks Ayrton became the first female fellow of the Royal Society for her work to improve the design of the electric arc lamps

that were starting to light up the world’s cities.

The female scientists, engineers, inventors, and doctors of the 19th century helped to start a revolution. By campaigning for women’s education and the opening up of careers in all fields to females, they paved the way for the increased rights women would attain in the 20th century, including the right to vote (see pp. 206–209).



**“THAT BRAIN OF
MINE IS SOMETHING
MORE THAN
MERELY MORTAL.”**

ADA LOVELACE, ENGLISH MATHEMATICIAN AND WRITER, 1843